

7.4 Good enough to eat (EAT)

This topic uses a case study of the production of sweets and biscuits:

- measuring and controlling the flow of a viscous liquid
- mechanical testing of products.

There are opportunities for students to develop practical skills and techniques and thus to carry out experimental and investigative activities.

Students will be assessed on their ability to:	Suggested experiments
18 understand and use the terms <i>density</i> , <i>laminar flow</i> , <i>streamline flow</i> , <i>terminal velocity</i> , <i>turbulent flow</i> , <i>upthrust</i> and <i>viscous drag</i> , for example, in transport design or in manufacturing	
19 recall, and use primary or secondary data to show that the rate of flow of a fluid is related to its viscosity	
20 recognise and use the expression for Stokes's Law, $F = 6\pi\eta r v$ and upthrust = weight of fluid displaced	
21 investigate, using primary or secondary data, and recall that the viscosities of most fluids change with temperature. Explain the importance of this for industrial applications	
25 investigate elastic and plastic deformation of a material and distinguish between them	
26 explore and explain what is meant by the terms <i>brittle</i> , <i>ductile</i> , <i>hard</i> , <i>malleable</i> , <i>stiff</i> and <i>tough</i> . Use these terms, give examples of materials exhibiting such properties and explain how these properties are used in a variety of applications, for example, safety clothing, foodstuffs	